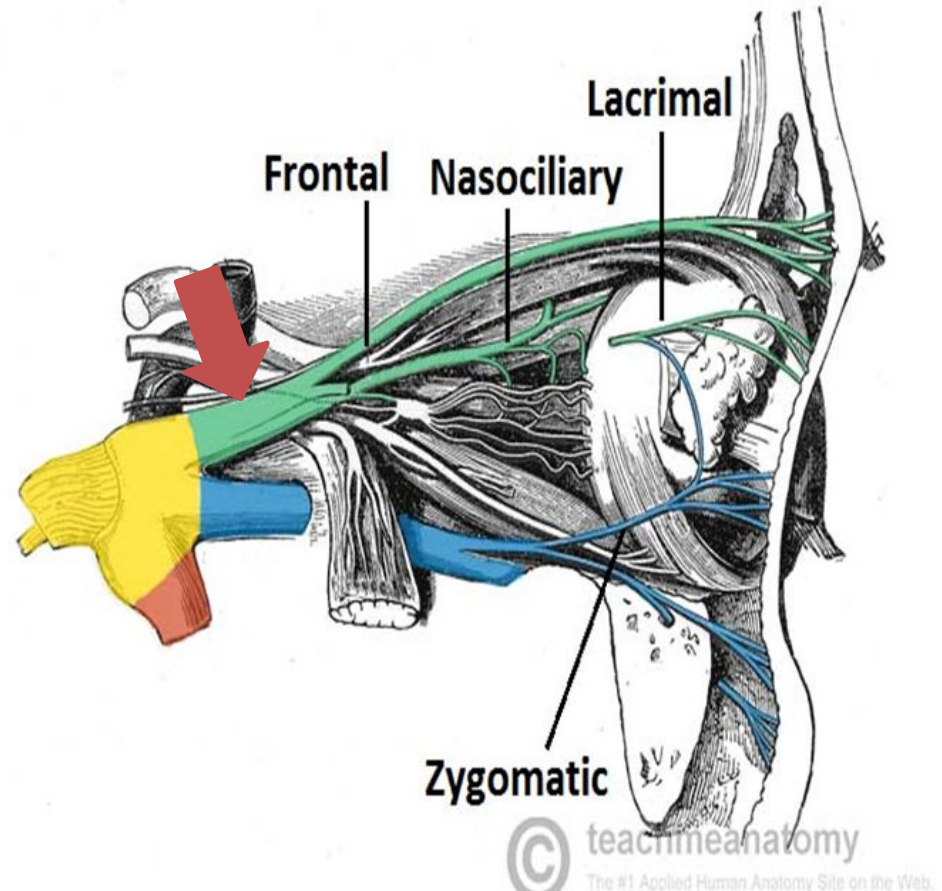


Ophthalmic nerve (V1):

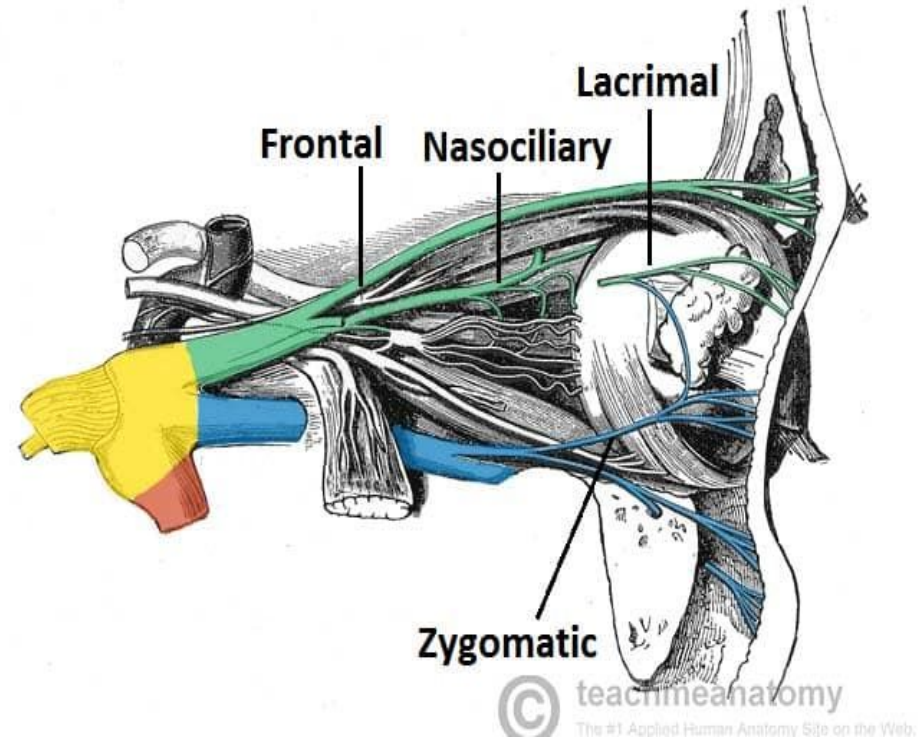
- The ophthalmic division is sensory
- It leaves the skull through superior orbital fissure to reach the orbit.
- It has three terminal branches viz, **frontal**, **lacrimal & nasociliary nerves**.



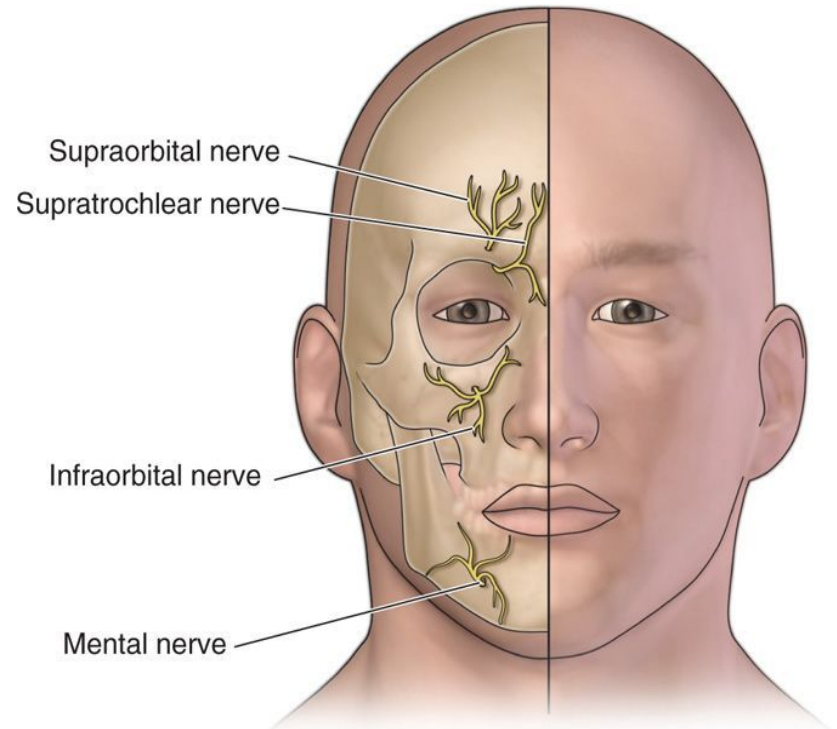
Branches of the Ophthalmic nerve:

1. Frontal nerve:

- Largest terminal branch of the ophthalmic nerve.
- Begins & runs in the lateral wall of the cavernous sinus.
- Enters the orbit through superior orbital fissure.
- Divides into the **smaller supratrochlear & larger supraorbital branches.**

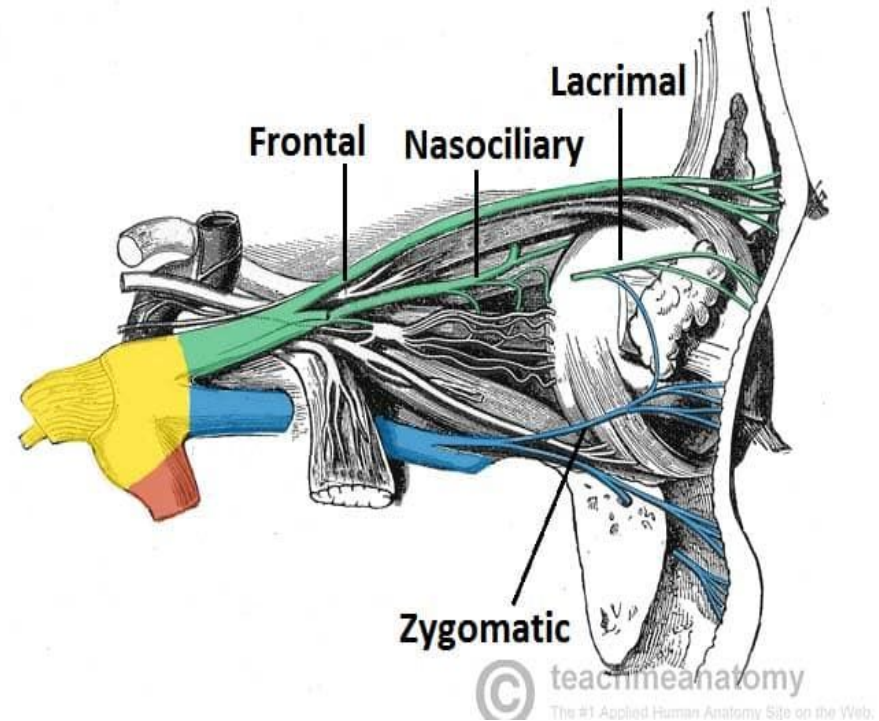


- i. **Supratrochlear nerve:** exits the orbit a little away from the median plane, and
 - Supplies the **conjunctiva**, the **upper eyelid**, & a **little part of the forehead above the root of the nose**.
- ii. **Supraorbital nerve:** exits the orbit through supraorbital foramen a little lateral to the supratrochlear nerve, and
 - Supplies the **conjunctiva**, **central part of upper eyelid**, **frontal sinus**, & **skin of forehead up to the vertex**.



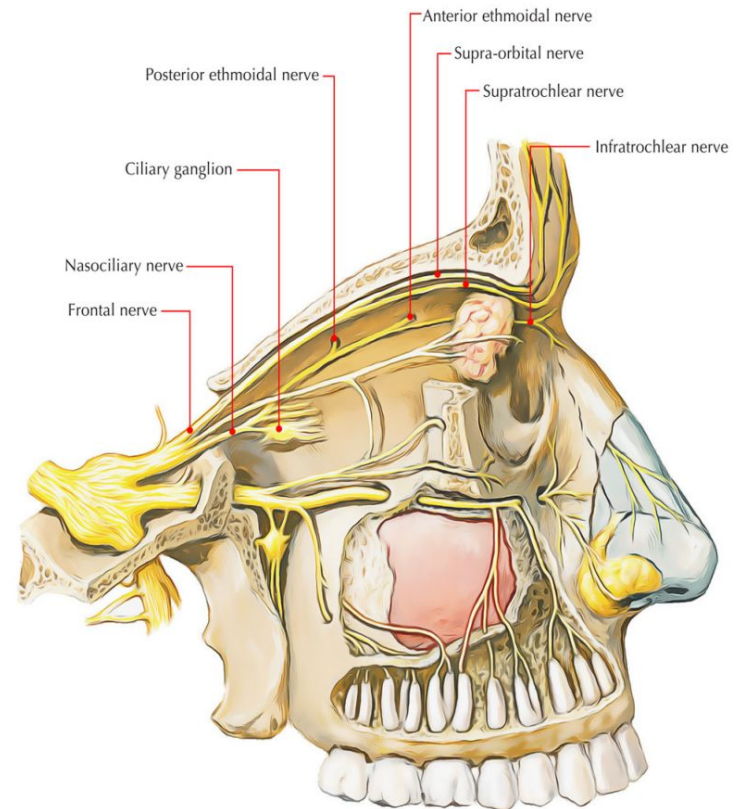
2. Lacrimal nerve:

- A smaller terminal branch of the ophthalmic nerve.
- Enters the orbit through superior orbital fissure.
- Runs along the upper part of the lateral rectus muscle & **ends by supplying the lacrimal gland** in the lateral part of upper eyelid.
- It carries **secretomotor fibers** to the lacrimal gland through its communication via zygomaticotemporal nerve.

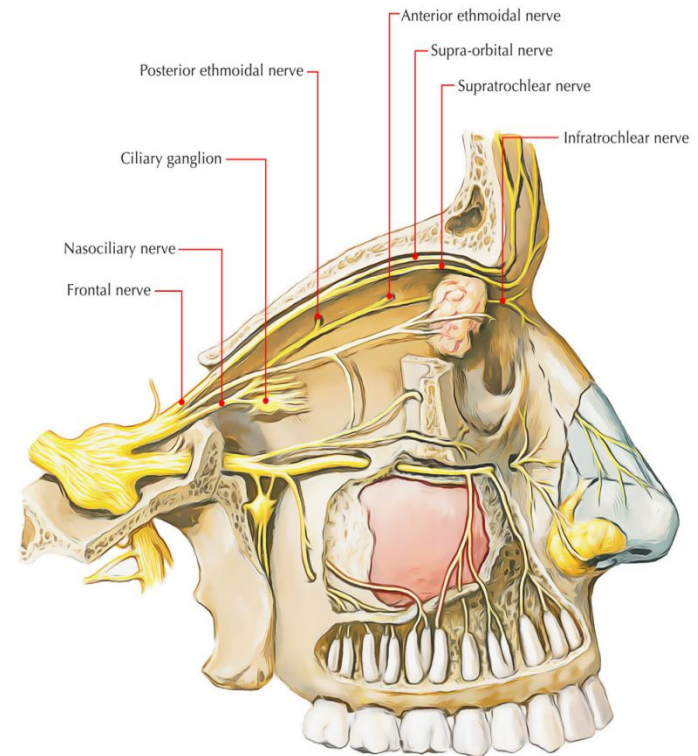


2. Nasociliary nerve:

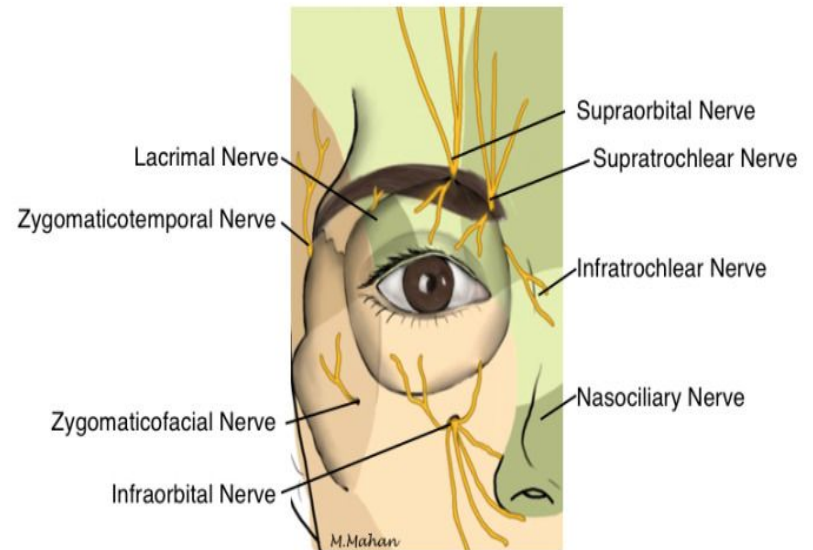
- Another terminal branch of the ophthalmic nerve.
- Enters the orbit through superior orbital fissure between the two divisions of oculomotor nerve.
- It passes above the optic nerve from lateral to medial side and ends by splitting into the **infratrochlear and anterior ethmoidal nerves**.



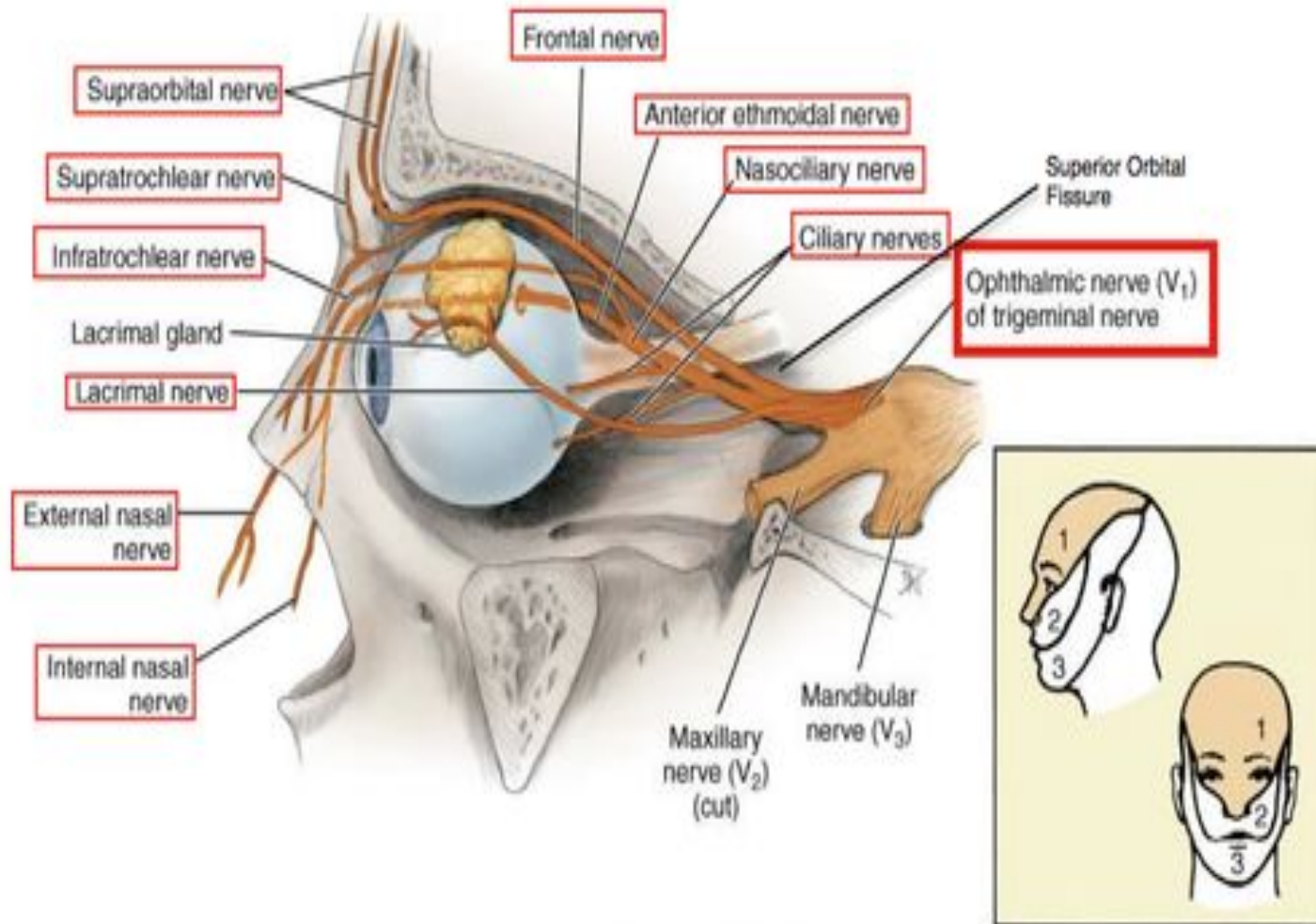
- The nasociliary nerve gives off:
 - i. **A communicating branch to the ciliary ganglion (sensory root) of the ganglion).**
 - ii. The **long ciliary nerves (2-3)** which give **sensory supply to the cornea, iris, & ciliary body** and **sympathetic supply to the dilator pupillae muscle.**
 - iii. The **posterior ethmoidal nerve** which supplies the **ethmoidal & sphenoidal air sinuses.**



- iv. The **infratrochlear nerve** which is smaller terminal branch and it emerges on the face above medial angle of eye to supply the **conjunctiva, lacrimal sac, medial parts of the eyelids, & upper half of the nose.**
- v. The **anterior ethmoidal nerve** which is a larger terminal branch and it exits the orbit through anterior ethmoidal foramen to supply the **mucosa of medial & lateral walls of the nose and skin of lower half of the nose via external nasal branch.**



Distribution of ophthalmic nerve:



REFERENCE HANDOUT: TRIGEMINAL NERVE BRANCHES (NOT INCLUDING HITCHHIKING PATHWAYS OF VII, IX)

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V1 Ophthalmic - GSA (Sensory) only - through Superior Orbital Fissure to Orbit

Nerve	Branches	Innervates
1. Frontal Nerve	a. Supraorbital Nerve	Scalp forehead, upper eyelid
	b. Supratrochlear Nerve	Scalp forehead, upper eyelid
2. Lacrimal Nerve		Upper eyelid
3. Nasociliary Nerve	a. Long Ciliary Nerve	Cornea of eye
	b. Ant. and Post. Ethmoidal Nerves	Nasal cavity, ethmoid sinus, tip of nose
	c. Infratrochlear Nerve	Upper eyelid, nose

V2 Maxillary - GSA (Sensory) only - through Foramen Rotundum to Pterygopalatine Fossa

Nerve	Branches	Innervates
1. Meningeal branches		Dura of mid. Cranial fossa
2. Ganglionic branches	a. Greater Palatine Nerve	Hard Palate
	b. Lesser Palatine Nerve	Soft Palate
	c. Nasopalatine Nerve	Nasal Cavity, Hard Palate
	d. Nasal branches	Nasal Cavity
3. Post. Sup. Alveolar Nerve		Maxillary teeth
4. Infraorbital nerve		Lower eyelid, nose, upper lip
	a. Ant. Sup. Alveolar Nerve	Maxillary teeth
	b. Mid. Sup. Alveolar Nerve	Maxillary teeth
5. Zygomatic nerve	a. Zygomaticofacial Nerve	Skin of cheek
	b. Zygomaticotemporal Nerve	Skin of temporal region

V3 Mandibular - GSA (Sensory) and SVE (Motor) - Foramen Ovale to Infratemporal Fossa

Nerve	Branches	Innervates
1. Nervous spinosus		Sensory to Dura of mid Cranial fossa
2. Motor branches		Motor to Med. Pterygoid, Tens. Tympani, Tensor Palati
3. Anterior division	a. Nerve to Lateral Pterygoid	Motor to Lateral Pterygoid
	b. Masseteric Nerve	Motor to Masseter
	c. Deep Temporal Nerve	Motor to Temporalis
	d. Buccal Nerve	Sensory to Cheek
4. Posterior Division	a. Auriculotemporal Nerve	Sensory to external auditory meatus, tympanic membrane, TMJ, lateral scalp
	b. Lingual Nerve	Sensory (touch) ant. 2/3 tongue
	c. Inferior Alveolar Nerve	Sensory to Mandibular teeth
	i. Nerve to Mylohyoid	Motor to Mylohyoid, ant. Digastric
	ii. Mental Nerve	Sensory to Chin, Lower lip